Priority-Aware Network Slicing for Disaster Environments Using Proximal Policy Optimization

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***Abstract*—In disaster scenarios, damaged base stations and backhaul links sharply reduce radio resources while emergency traffic surges, making resource allocation that guarantees the quality of service (QoS) of mission-critical communications essential.In this paper, inspired by the priority mechanisms in the Third Generation Partnership Project (3GPP) specifications, we define three user priority tiers (T0: general, T1: responder, T2: mission-critical command) and propose a proximal policy optimization (PPO)-based resource allocation scheme that optimizes the satisfaction of delay and reliability targets.The agent allocates resource block (RB) shares to nine classes formed by three slices and three tiers, and we conduct a tier on/off ablation over five training seeds by varying only the tier weights in the reward. During the disaster window, enabling tier priority (Tier-ON) increases the average QoS satisfaction of T2 by 54.9 percent and decreases that of T0 by 31.2 percent in relative terms, as the RB share of T2 rises from 0.171 to 0.404 while every active T0 class keeps a guaranteed minimum RB share.These results show that tier-aware PPO enables effective priority control in disaster scenarios without starving lower-tier users.**

***Index Terms*—Network slicing, disaster communications, priority tiers, deep reinforcement learning, PPO.**

I. Introduction

In disaster scenarios, damage to base stations and backhaul links reduces the available resources while the traffic of rescue personnel surges, so that the supply falls short of the demand. If all traffic is treated equally, mission-critical communications must compete with general traffic and may suffer excessive delay or loss.

The Third Generation Partnership Project (3GPP) already specifies priority handling for emergency communications in its fifth-generation (5G) standards. For example, the Allocation and Retention Priority (ARP) determines which quality of service (QoS) flows are admitted first and which flows can preempt others when resources are limited [1]. The Multimedia Priority Service (MPS) gives priority users preferential access to network resources, while allowing general users to retain network access subject to regulatory policy [2]. 3GPP also specifies that mission-critical services for public safety require low latency, high availability, and high reliability [3]. However, xSlice [4], a representative learning-based slicing scheme, applies the same QoS demands and regret weights to all sessions within a slice, and thus cannot distinguish the mission criticality of users within the same slice.

For example, if a rescue drone streaming video from collapsed debris and a citizen making a video call from a shelter share the same enhanced mobile broadband (eMBB) slice, the conventional approach cannot differentiate them, and the rescue communication may be delayed.

In this paper, we propose a resource allocation scheme based on proximal policy optimization (PPO) [5] with tier-based intra-slice priority, and we compare the QoS with and without tier priority. Since the agent directly determines the resource block (RB) shares of nine slice-tier classes, the tier ordering is learned under tier-weighted rewards rather than enforced by fixed rules. The main contributions of this paper are as follows.

- We formulate a disaster-aware Markov decision process (MDP) that includes the load of each slice-tier class in the state and slice-tier RB shares in the action.
- We show the trade-off in which tier priority improves the QoS of T2 at the cost of T0, while a minimum-share floor keeps every active T0 class served and RBs are dynamically reallocated during the disaster window.

II. System Model

A. Network and Priority Tiers

We consider a single cell with N_{RB} resource blocks and three slices $s \in \mathcal{S} = \{\mathrm{eMBB}, \mathrm{URLLC}, \mathrm{mMTC}\}$, where ultra-reliable low-latency communication (URLLC) and massive machine-type communication (mMTC) are served together with eMBB. Each slice s has its own delay target D_s and block error rate (BLER) target B_s. Within each slice, users are divided into three priority tiers $k \in \mathcal{K} = \{0, 1, 2\}$, as summarized in Table I, where a higher index indicates a higher priority. The tiers are logical layers defined in this work, and tier membership is assigned according to the mission criticality of each incident. T2 groups command traffic and autonomous devices because both generate delay-sensitive control traffic. Each pair $c = (s, k)$ forms a class with $n_c(t)$ active sessions at step t, which yields nine classes in total.

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This work was supported in part by the Dongguk University Research Fund of 2026, in part by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. RS-2026-25481311), and in part by Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. RS-2026-25523610).

TABLE I
DEFINITION OF PRIORITY TIERS

Tier	Users	Weight[a]	Arrival[b]
T0	General public	1	1.2 / 1.2
T1	Field responders	3	0.2 / 0.8
T2	Command, autonomous	8	0.05 / 0.25[c]

[a]Tier-ON. [b]Poisson mean per slice per step (normal / disaster).
[c]Burst probability; each burst adds Poisson(1.2) sessions per slice.

B. Disaster Traffic Model

Each episode lasts T steps and contains a disaster window. New sessions arrive according to Table I, so that T1 traffic increases and T2 bursts occur more frequently during the disaster window. Each session of tier k departs with probability $p_k\,(\beta + (1-\beta)q_c)$, where q_c is the QoS satisfaction defined in (3) and $\beta = 0.25$. Therefore, poorly served sessions stay longer, and the allocation affects the future load. The fraction of operational cell capacity $\eta(t)$ decreases by a random amount in $[0, 0.045]$ per step down to 0.4 during the disaster window, and it recovers by 0.04 per step otherwise.

III. PRIORITY-AWARE DRL FRAMEWORK

The agent interacts with the cell environment at each step, and it is trained to maximize the expected discounted return $\mathbb{E}\left[\sum_t \gamma^t R_t\right]$. The key elements of the framework are as follows.

A. State

The state is a 39-dimensional vector in $[0, 1]$:

$$\mathbf{o}_t = \left[\boldsymbol{\rho}_t,\ \mathbf{u}^d_{t-1},\ \mathbf{u}^b_{t-1},\ \boldsymbol{\phi}_{t-1},\ \eta(t),\ \delta(t),\ t/T\right], \quad (1)$$

where $\boldsymbol{\rho}_t = \mathbf{n}_t / N_{\max}$ is the load of the nine classes, $\mathbf{u}^d_{t-1}$ and $\mathbf{u}^b_{t-1}$ are the delay and BLER satisfaction of the previous step, $\boldsymbol{\phi}_{t-1}$ is the previous allocation, and $\delta(t)$ indicates the disaster window.

B. Action

The action $\mathbf{a}_t \in [0, 1]^9$ is a preference vector over the nine classes. Let $\mathcal{A}_t$ be the set of active classes. The RB share of an active class is

$$\phi_c(t) = f + (1 - |\mathcal{A}_t| f)\,\frac{a_c}{\sum_{c' \in \mathcal{A}_t} a_{c'}}, \quad (2)$$

where $f = \min(0.03,\ 0.9/|\mathcal{A}_t|)$ is a minimum-share floor, and $\phi_c(t) = 0$ for inactive classes. The floor guarantees that no active class is starved, while the remaining RBs are distributed by the learned policy.

C. Reward

The sessions in each class share their RBs equally, and the delay d_c and BLER ϵ_c of each session are computed from its allocated RBs using a simplified analytical model, where both decrease as more RBs are allocated. The QoS satisfaction of class c is the average of its delay and BLER satisfaction:

$$q_c = \frac{1}{2}\left[\min\left(1,\ \frac{D_s}{d_c}\right) + \min\left(1,\ \frac{B_s}{\epsilon_c}\right)\right]. \quad (3)$$

TABLE II
SUMMARY OF SIMULATION SETTINGS

Description	Value
Number of RBs (N_{RB})	60
Capacity per RB	0.7 Mbps
Maximum sessions per class ($N_{\max}$)	20
Episode length (T)	100 steps
Disaster window	Steps 30 to 75
Delay target (D_s)	25, 8, 80 ms
BLER target (B_s)	0.02, 0.005, 0.04
Delay model (processing, load)	2 ms, 20 ms·Mbps
BLER model (base, reference rate)	0.08, 0.5 Mbps
Minimum rate	0.02 Mbps
Departure probability (p_k)	0.16, 0.12, 0.09
Hidden layers (actor and critic)	256, 256
Parallel environments	8
Rollout length / batch size	512 / 256
Epochs per update	20
Discount factor (γ)	0.99
GAE factor	0.95
Clipping range	0.2
Learning rate	3×10^{-4}, linear decay
Training timesteps	10^6
Training seeds / test episodes	5 / 100

D_s, B_s: eMBB, URLLC, mMTC. p_k: T0, T1, T2.

The reward is the tier-weighted average QoS satisfaction over all sessions:

$$R_t = \frac{\sum_c w_k\, n_c\, q_c}{\sum_c w_k\, n_c}, \quad (4)$$

where w_k is the weight of tier k. Tier-ON uses $\mathbf{w} = [1, 3, 8]$, and Tier-OFF uses $\mathbf{w} = [1, 1, 1]$. The tier weight works as a soft priority, whereas the minimum-share floor in (2) works as a hard guarantee.

D. Training Algorithm

The agent is trained with PPO, which uses an actor network to output the action and a critic network to estimate the state value. The advantage is computed by generalized advantage estimation (GAE) [6], and the policy update is clipped to keep training stable.

IV. SIMULATION RESULTS

Table II summarizes the simulation settings. We use the PPO implementation of Stable-Baselines3 [7]. The environment and traffic model are identical for Tier-ON and Tier-OFF, and only the tier weights differ. Each setting is trained with five seeds and evaluated with deterministic policies on 100 unseen test episodes.

A. Training Convergence of PPO

Fig. 1 shows the training curves of both settings over 10^6 timesteps. Since Tier-ON and Tier-OFF are driven by different reward weights, their curves are not directly comparable, and each setting is therefore assessed against the random and uniform allocation baselines evaluated under the same reward. In both settings, the reward increases rapidly in the early phase, exceeds both baselines within the first half of training, and subsequently plateaus, which indicates that the learned

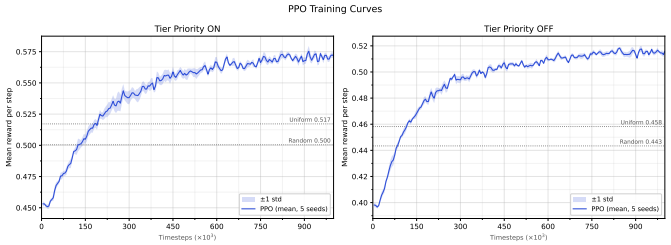


Fig. 1. Training curves of PPO with tier priority ON and OFF (mean over five training seeds; shaded areas indicate ± 1 standard deviation; dotted lines indicate the random and uniform allocation baselines).

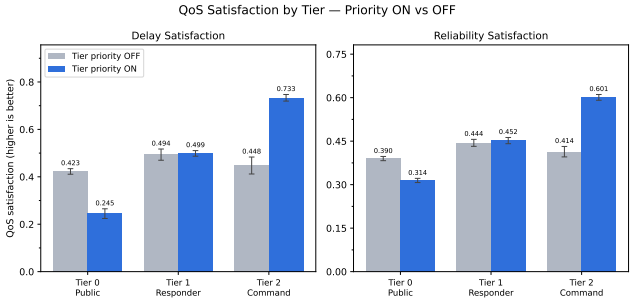


Fig. 2. QoS satisfaction of each tier during the disaster window (mean over five training seeds; error bars indicate ± 1 standard deviation).

policy has converged. The shaded regions remain narrow over the entire run, showing that the five seeds follow nearly identical trajectories and that the training is reproducible.

B. QoS Performance by Tier

Fig. 2 shows the QoS satisfaction of each tier during the disaster window. Under Tier-ON, both the delay and the reliability satisfaction of T2 rise well above those of the remaining tiers, whereas T0 falls below its Tier-OFF level and T1 remains nearly unchanged. Averaged over the two metrics, the satisfaction of T2 increases by 54.9%, whereas that of T0 decreases by 31.2%. This result confirms that the tier weights shift resources toward the highest tier, and that prioritizing one tier inevitably costs the others under resource scarcity.

Table III reports the average RB share of each tier during the disaster window, which accounts for the above trend. Tier-ON more than doubles the RB share of T2 and reduces that of T0 to less than half, while the share of T1 remains unchanged. Although T0 forfeits most of its share, the minimum-share floor keeps every active T0 class served. The share of T1 is preserved because the tier weights hardly alter its contribution to the weighted reward. Consequently, resources are mainly reallocated from T0 to T2, and the communications of field responders are protected even during T2 surges.

V. Conclusion

We proposed a tier-aware PPO-based resource allocation scheme for disaster scenarios, in which three logical priority tiers inspired by 3GPP priority mechanisms are applied within network slices. The simulation results showed that tier priority

TABLE III
Average RB Share of Each Tier During the Disaster Window

Tier	Tier-OFF	Tier-ON
T0	0.442	0.209 ↓
T1	0.387	0.387 –
T2	0.171	0.404 ↑

improves the delay and reliability satisfaction of T2 by 54.9% at the cost of a 31.2% reduction for T0, by more than doubling the RB share of T2 while a minimum-share floor keeps every active T0 class served. Future work includes adaptive tier weights after the disaster window, multi-cell scenarios, and validation on an open radio access network (O-RAN) testbed.

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